E-Commerce Sales Data Analysis and Business Intelligence Dashboard

SWYNEX Technologies – Data Analytics Internship
Final Project Report – 13-10-2026

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1. Project Overview

This project focuses on analyzing an e-commerce sales dataset using Python and Power BI. It combines data cleaning, exploratory data analysis, and interactive dashboard development to identify sales trends, customer preferences, operational performance, and business improvement opportunities.

The project was completed in three main stages:

- Task 1: Data Cleaning and Preparation using Python
- Task 2: Exploratory Data Analysis (EDA)
- Task 3: Interactive Dashboard Development using Power BI

2. Problem Statement

E-commerce businesses generate large volumes of transactional data. However, raw data may contain inconsistencies, missing values, duplicates, and formatting issues that can affect reporting accuracy.

The objective of this project is to transform raw sales data into reliable information and answer important business questions:

- Which product category generates the highest revenue?
- Which region performs best in terms of sales?
- How does revenue change over time?
- Which payment method is most frequently used?
- How efficiently are orders delivered?
- What do customer ratings indicate about overall customer satisfaction?

3. Dataset Information

The dataset contains 5,000 e-commerce transaction records and 12 columns.

Dataset attributes:

Column	Description
Order_id	Unique order identifier
Order_date	Date of the order
Customer_id	Customer identifier
Product_category	Product category
Region	Sales region
Quantity	Quantity purchased
Unit_price	Price per unit
Discount	Discount applied
Payment_method	Mode of payment
Delivery_days	Delivery duration
Customer_rating	Customer rating
Revenue	Revenue Generated

```
# Display first 10 rows
display(df.head(10))

# Display dataset dimensions
print("Total Rows:", df.shape[0])
print("Total Columns:", df.shape[1])
```

[6] ✓ 0.0s

	order_id	order_date	customer_id	product_category	region	quantity	unit_price	discount	payment_method	delivery_days	customer_rating	revenue
0	10001	01-01-2022	1102	Beauty	South	7	373.65	0.28	Wallet	10	4.7	1883.20
1	10002	01-02-2022	1435	Clothing	South	7	47.74	0.09	Card	6	3.9	304.10
2	10003	01-03-2022	1860	Beauty	East	3	311.28	0.31	COD	6	2.5	644.35
3	10004	01-04-2022	1270	Electronics	West	5	524.47	0.02	Wallet	6	1.6	2569.90
4	10005	01-05-2022	1106	Clothing	West	5	139.87	0.33	Wallet	4	4.9	468.56
5	10006	01-06-2022	1071	Electronics	West	2	264.44	0.20	Card	3	3.3	423.10
6	10007	01-07-2022	1700	Clothing	South	6	564.60	0.23	Card	1	1.8	2608.45
7	10008	01-08-2022	1020	Beauty	North	6	265.71	0.13	Card	2	3.7	1387.01
8	10009	01-09-2022	1614	Clothing	South	6	224.65	0.27	Card	2	1.4	983.97
9	10010	01-10-2022	1121	Clothing	West	4	272.71	0.06	Card	10	4.5	1025.39

```
Total Rows: 5000
Total Columns: 12
```

Outcome: The dataset consists of 5,000 rows and 12 columns containing e-commerce transaction details. The screenshot displays a preview of the first 10 records along with the dataset dimensions. The complete dataset was used for data cleaning, exploratory data analysis, and dashboard development.

4. Data Cleaning and Preparation

Python and Pandas were used to inspect, clean, and prepare the raw dataset for further analysis.

The data-cleaning process included:

- Checking missing values and duplicate records.
- Identifying invalid and inconsistent values.
- Standardizing the order date format.
- Reviewing numerical and categorical columns.
- Validating the dataset after cleaning.
- Exporting the prepared dataset as a CSV file.

```
print("Missing Values:")
print(df.isnull().sum())

print("Duplicate Rows:", df.duplicated().sum())

print("Duplicate Order IDs:", df["order_id"].duplicated().sum())

print("\nData Types:")
print(df.dtypes)
```

✓ 0.0s

```
Missing Values:
order_id            0
order_date          0
customer_id         0
product_category    0
region              0
quantity            0
unit_price          0
discount            0
payment_method      0
delivery_days       0
customer_rating     0
revenue             0
dtype: int64
Duplicate Rows: 0
Duplicate Order IDs: 0

Data Types:
order_id                     int64
order_date          datetime64[ns]
customer_id                  int64
product_category            object
region                      object
quantity                     int64
unit_price                 float64
...
delivery_days                int64
customer_rating            float64
revenue                    float64
dtype: object
```

Outcome: A cleaned dataset containing 5,000 records was prepared for EDA and Power BI visualization.

5. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed using Python to understand the dataset and identify meaningful patterns.

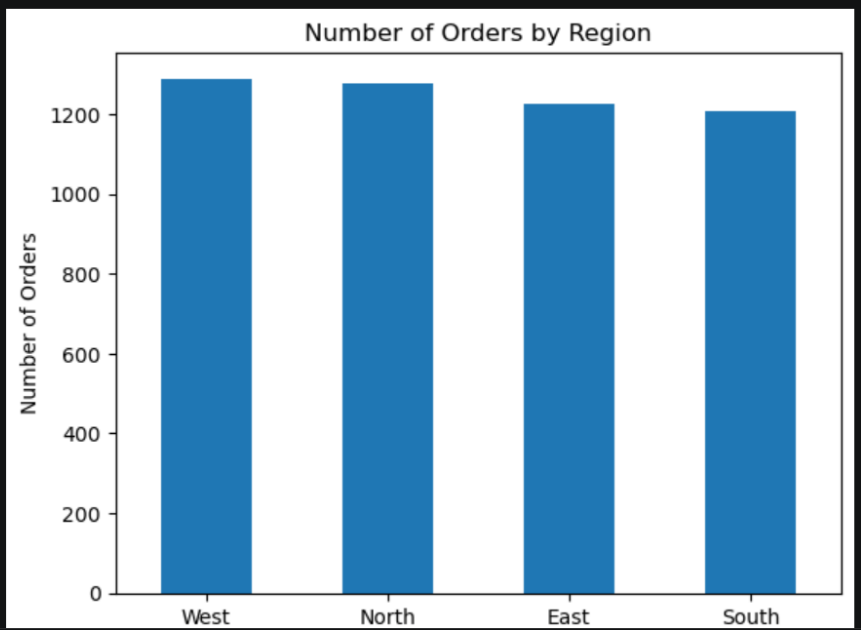
The analysis covered:

- Descriptive statistics.

```
region_orders.plot(kind="bar")

plt.title("Number of Orders by Region")
plt.xlabel("Region")
plt.ylabel("Number of Orders")
plt.xticks(rotation=0)
plt.show()
```

✓ 0.1s

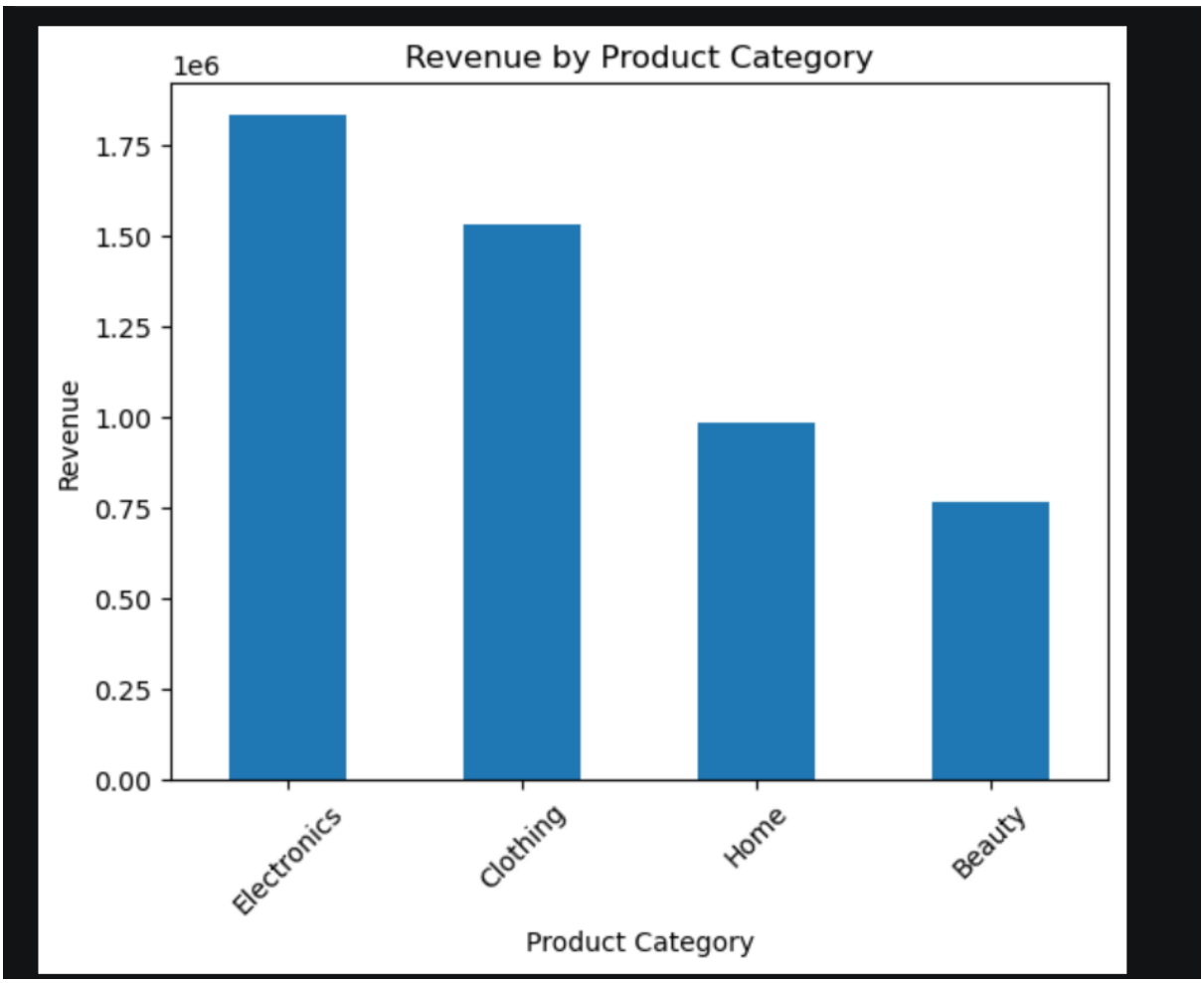


- Product category performance

```
#chart
category_revenue.plot(kind="bar")

plt.title("Revenue by Product Category")
plt.xlabel("Product Category")
plt.ylabel("Revenue")
plt.xticks(rotation=45)
plt.show()
```

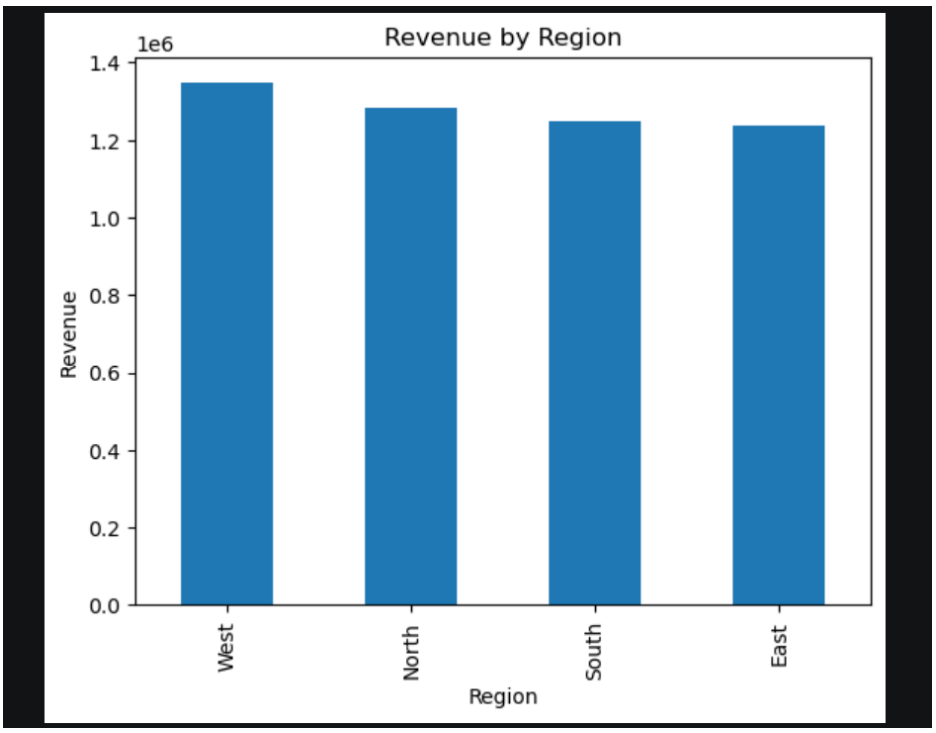
e.



- Regional sales performance.

```
region_revenue.plot(kind="bar")

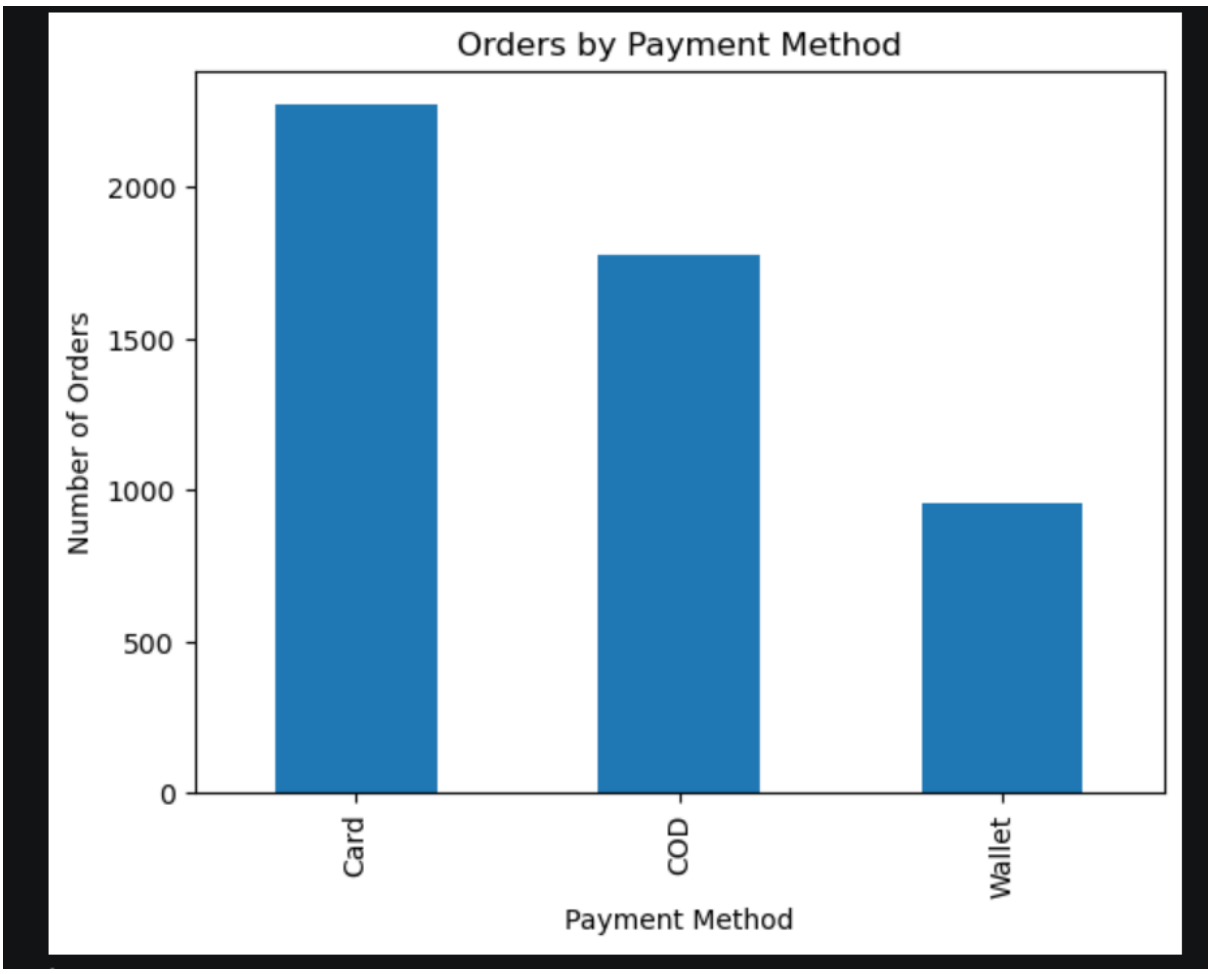
plt.title("Revenue by Region")
plt.xlabel("Region")
plt.ylabel("Revenue")
plt.show()
```



- Payment method preferences.

```
#chart
payment_usage.plot(kind="bar")

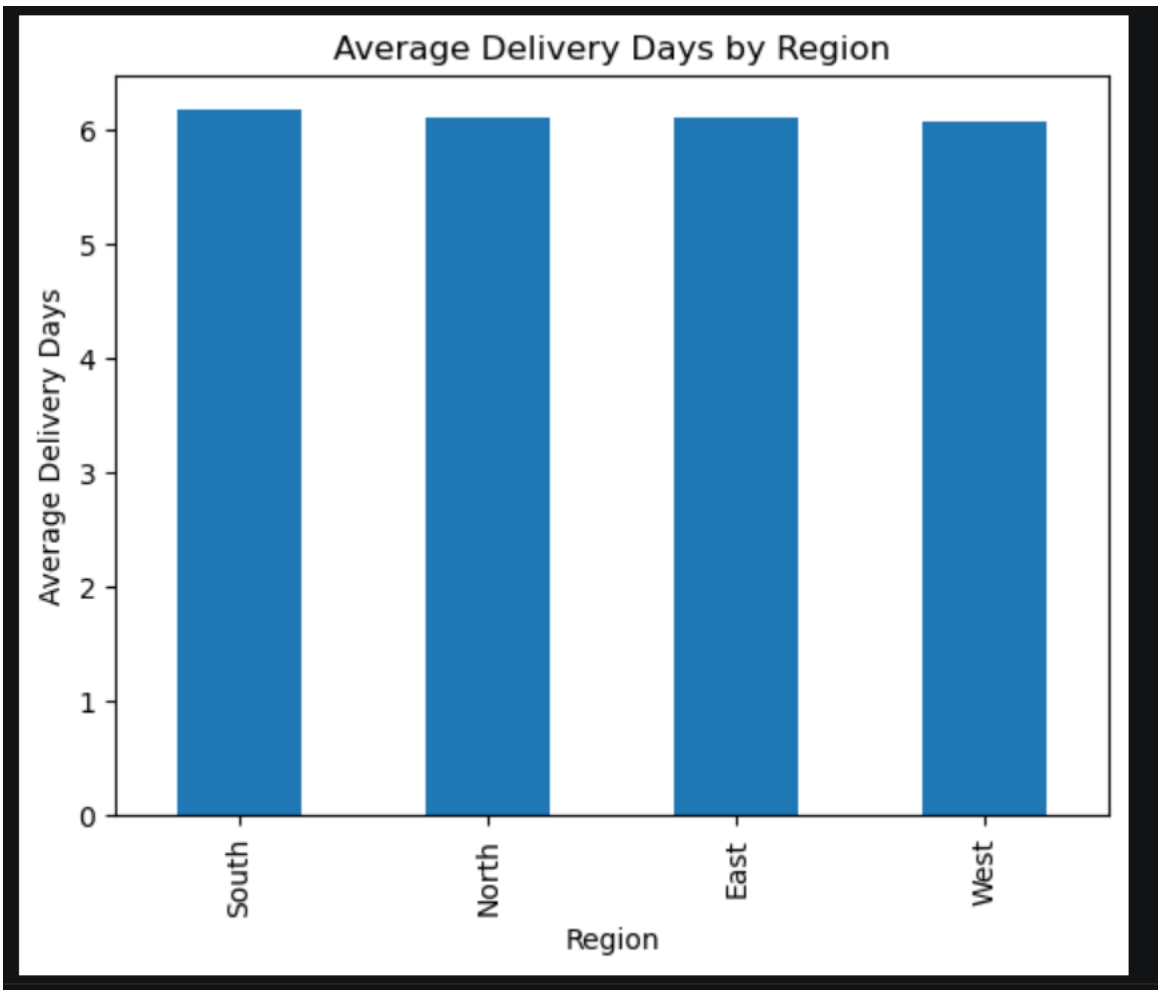
plt.title("Orders by Payment Method")
plt.xlabel("Payment Method")
plt.ylabel("Number of Orders")
plt.show()
```



- Delivery performance

```
#chart
delivery_region.plot(kind="bar")

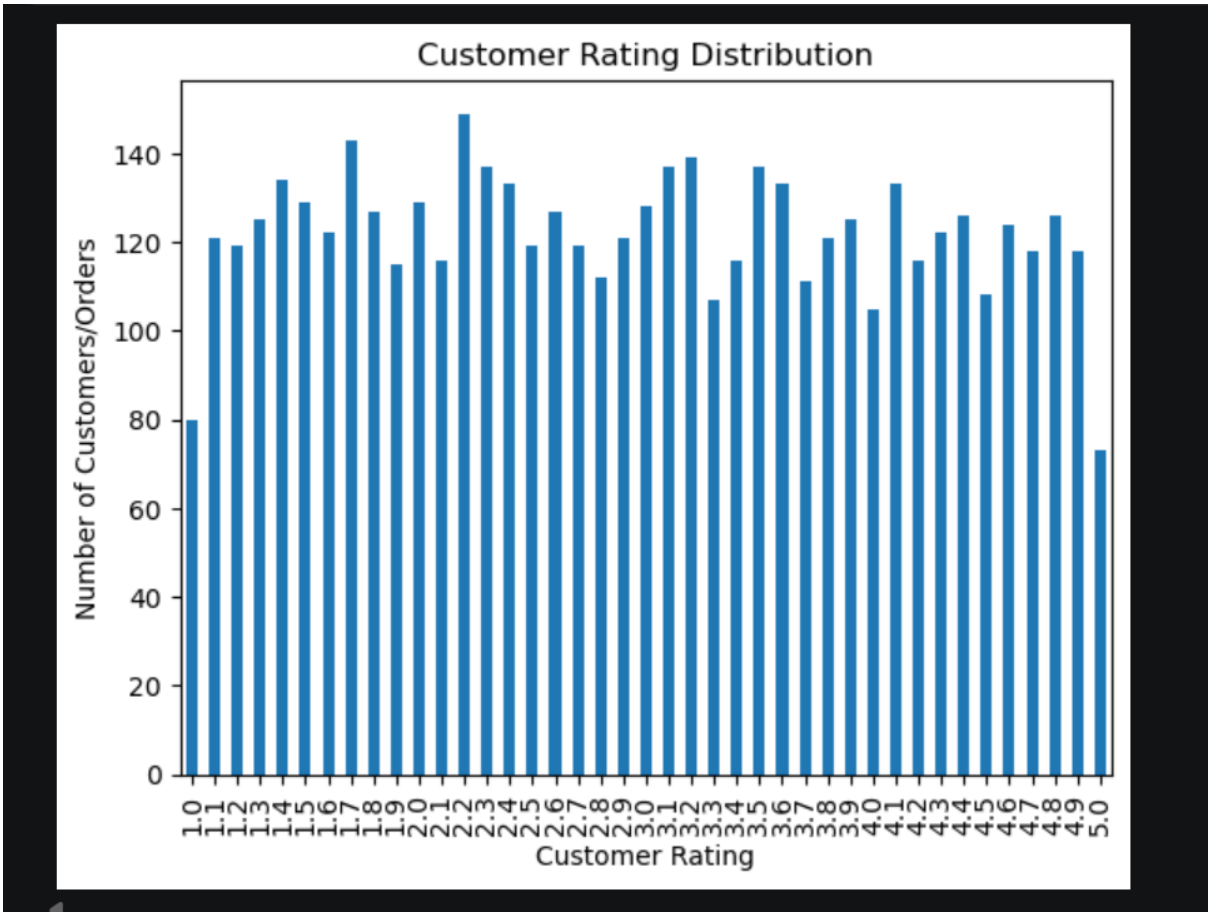
plt.title("Average Delivery Days by Region")
plt.xlabel("Region")
plt.ylabel("Average Delivery Days")
plt.show()
```



- Customer rating analysis.

```
#chart
rating_counts.plot(kind="bar")

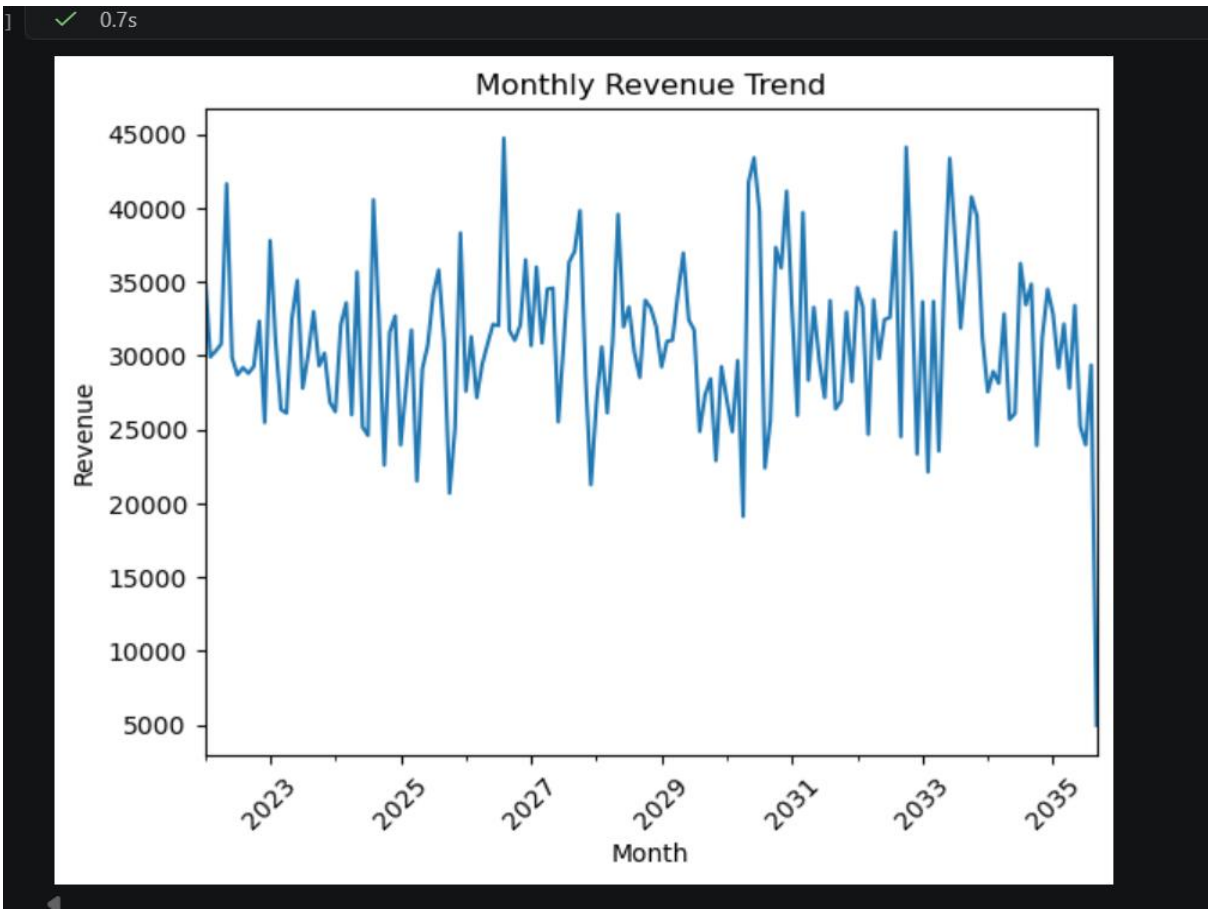
plt.title("Customer Rating Distribution")
plt.xlabel("Customer Rating")
plt.ylabel("Number of Customers/Orders")
plt.show()
```



- Revenue trends over time.

```
#create a line chart
monthly_revenue.plot(kind="line")

plt.title("Monthly Revenue Trend")
plt.xlabel("Month")
plt.ylabel("Revenue")
plt.xticks(rotation=45)
plt.show()
```



- Revenue distribution and outlier detection.

```
Q1 = df["revenue"].quantile(0.25)
Q3 = df["revenue"].quantile(0.75)

IQR = Q3 - Q1

lower_limit = Q1 - 1.5 * IQR
upper_limit = Q3 + 1.5 * IQR

anomalies = df[
    (df["revenue"] < lower_limit) |
    (df["revenue"] > upper_limit)
]

print("Number of revenue anomalies:", len(anomalies))
print(anomalies)
```

0.0s

```
Number of revenue anomalies: 67
```

Using the Interquartile Range (IQR) method, 67 potential revenue outliers were identified for further investigation.

6. Power BI Dashboard Development

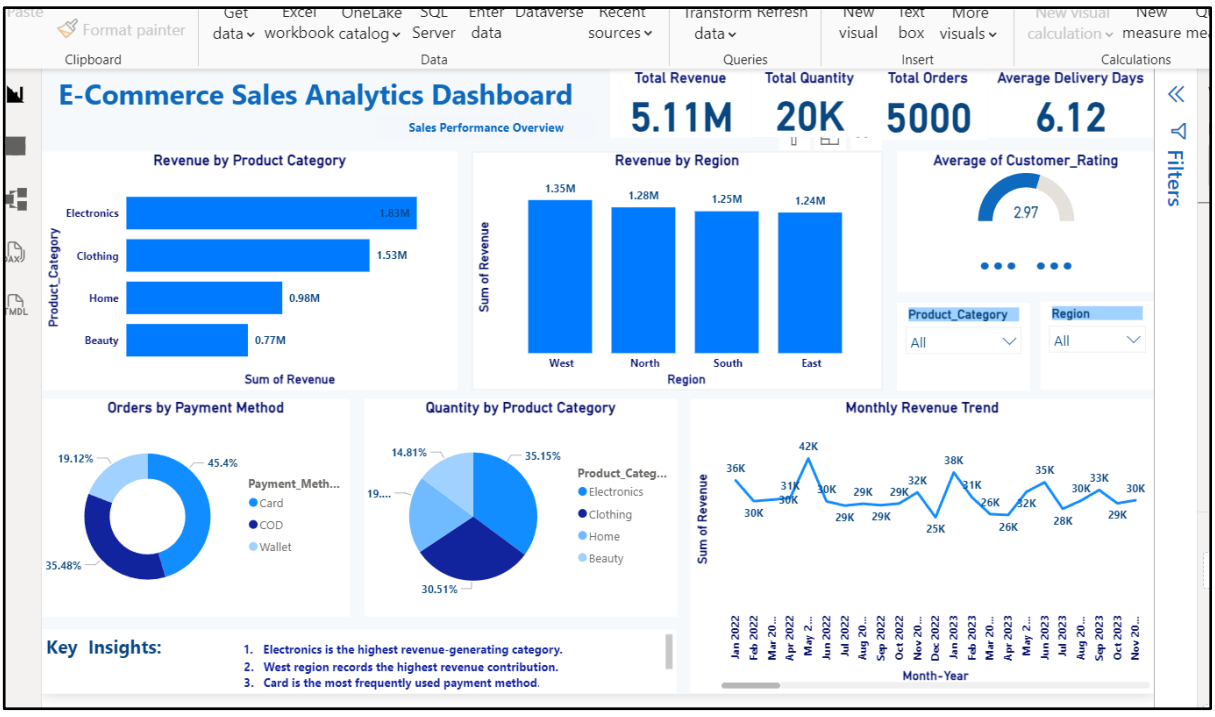
An interactive E-Commerce Sales Analytics Dashboard was created using the cleaned dataset.

Dashboard KPIs:

- Total Revenue: Approximately 5.11M
- Total Orders: 5,000
- Total Quantity: Approximately 20K
- Average Delivery Days: 6.12
- Average Customer Rating: 2.97/5

Dashboard Visualizations:

- Revenue by Product Category
- Revenue by Region
- Monthly Revenue Trend
- Orders by Payment Method
- Quantity by Product Category
- Interactive filters and slicers



The dashboard provides an overview of business performance and allows users to explore different sales dimensions.

7. Key Business Insights

The following insights were identified during the analysis:

1. **Electronics** generated the highest total revenue among product categories.
2. The **West region** recorded the highest revenue, while the East region recorded the lowest.
3. **Card** was the most frequently used payment method.
4. The **South region** had the highest average delivery time, approximately 6.17 days.
5. The overall average customer rating was **2.97 out of 5**, indicating an opportunity to improve customer satisfaction.
6. IQR analysis identified **67 potential revenue outliers**, which may represent unusually high-value transactions.
7. Revenue analysis revealed fluctuations across different years, highlighting the importance of monitoring sales trends.

8. Business Recommendations

Based on the findings, the following actions are recommended:

- Investigate the factors contributing to strong Electronics sales and identify opportunities to improve other product categories.
- Analyze the performance gap between the West and East regions.
- Review delivery operations, particularly in the South region.
- Investigate customer feedback to identify potential reasons for moderate customer ratings.
- Monitor revenue outliers to distinguish genuine high-value transactions from possible data issues.
- Use the Power BI dashboard for ongoing performance monitoring and reporting.

9. Conclusion

This project demonstrates the complete data analytics workflow, from preparing raw data to communicating business insights through an interactive dashboard.

By combining Python, Pandas, exploratory analysis, Power BI, and DAX, the project transformed transactional data into meaningful visual reports that can support business decision-making.

Completing this project strengthened my practical skills in data cleaning, analytical thinking, data visualization, and business intelligence reporting.